Bilinear and Additive Attention Scoring

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Where the scoring function fits

At target step t, attention first computes a raw relevance score for each source position j:

$$e_{tj} = a(s_{t-1}, h_j)$$

Mental model

decoder need s_{t-1} $+$ source representation h_j $\longrightarrow$ score e_{tj}

- s_{t-1}: what the decoder currently knows and what it seems to need next.
- h_j: contextual representation of source word j.
- e_{tj}: a raw compatibility score, not yet a probability.
- Softmax across all j turns these scores into attention weights α_{tj}.

Bilinear scoring: formula and dimensions

$$e_{tj} = s_{t-1}^{\top} W_a h_j$$

Dimension check

Assume

$$s_{t-1} \in \mathbb{R}^{d_s}, \qquad h_j \in \mathbb{R}^{d_h}$$

Then choose

$$W_a \in \mathbb{R}^{d_s \times d_h}$$

so that

$$(1 \times d_s)(d_s \times d_h)(d_h \times 1) = 1$$

The result is a scalar.

Step by step

1. Transform the source state: $W_a h_j$
2. Compare it with the decoder state: $s_{t-1}^{\top}(W_a h_j)$
3. Output one number: the relevance score

Bilinear scoring: three useful ways to read it

1. Learned metric

The dot product $s_{t-1}^\top h_j$ compares the vectors as they are.
The bilinear form says: first learn a matrix W_a then compare.
So the model learns a better notion of similarity than the raw dot product.

2. Equivalent view

Because matrix multiplication is associative,

$$s_{t-1}^\top W_a h_j = (W_a^\top s_{t-1})^\top h_j$$

You can think of it as transforming either side

3. Coordinate expansion

$$e_{tj} = \sum_{m=1}^{d_s} \sum_{n=1}^{d_h} s_{t-1,m} \, W_{a,mn} \, h_{j,n}$$

Each decoder feature can interact with each encoder feature, weighted by a learned parameter.

Key limitation

It is still linear in each argument. That makes it efficient and clean, but less flexible than a nonlinear scorer.

Additive scoring: formula and dimensions

$$e_{tj} = v_a^\top \tanh(W_s s_{t-1} + W_h h_j + b_a)$$

Dimension check

Let the alignment hidden layer have size d_a.

$$W_s \in \mathbb{R}^{d_a \times d_s}, \qquad W_h \in \mathbb{R}^{d_a \times d_h}$$

$$v_a \in \mathbb{R}^{d_a}, \qquad b_a \in \mathbb{R}^{d_a}$$

Then $W_s s_{t-1}$ and $W_h h_j$ both live in the same d_a-dimensional alignment space.

Why "additive"?

Because the projected states are combined by addition:

$$W_s s_{t-1} + W_h h_j + b_a$$

After that, a nonlinearity and a final vector v_a turn the hidden representation into one scalar score.

Additive scoring: a tiny neural network for compatibility

Step-by-step computation

$$\underbrace{u_s = W_s s_{t-1}}_{\text{project decoder state}} \qquad \underbrace{u_h = W_h h_j}_{\text{project encoder state}}$$

$$\underbrace{z_{tj} = u_s + u_h + b_a}_{\text{combine in shared space}} \qquad \underbrace{g_{tj} = \tanh(z_{tj})}_{\text{nonlinear hidden features}}$$

$$\underbrace{e_{tj} = v_a^\top g_{tj}}_{\text{collapse to one score}}$$

Interpretation

Additive attention acts like a small feed-forward network that judges how well the decoder state matches one source position.

Hidden units learn compatibility features; $\tanh$ makes the score nonlinear.

Bilinear vs. additive: the takeaway

Method	What it learns	Best intuition
Bilinear	A matrix W_a defining a learned linear comparison	"Learn a metric, then measure similarity."
Additive	Two projections, a hidden layer, a nonlinearity, and a final scoring vector	"Feed the pair into a tiny neural scorer."

One-sentence summary

Bilinear attention is a learned linear comparison; additive attention is a learned nonlinear compatibility network.

After either scorer produces e_{tj} for every source position j, softmax turns those scores into attention weights.